

Random Variables and Expectations

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CS 0220 2022

April 8, 2022

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Numerical Values of Outcomes

Sometimes it makes sense to attach a numerical value to each outcome of a probability space.

Example: We ask people to name all the planets they can.

- Earth, Mars
- Earth, Neptune, Mars, Saturn
- Mercury, Venus, Earth, Mars, Jupiter, Saturn, Neptune, Uranus
- Mars, Jupiter

How many outcomes? $2^8 = 256$, assuming we ignore any fictional planets they name like “Tatooine” or “Pluto”.

If we want to summarize the results, we might assign each outcome a *statistic*, that is, a numerical summary. A natural choice is the number of planets they named: 2, 4, 8, 2.

Random variable

Definition: A *random variable* R on a probability space is a function whose domain is the sample space.

Example: Let's say the sample space is a deck of cards and R maps a number card to its value and a face card to 10 and ace to 1. So, $R(2♥) = 2$ and $R(J♣) = 10$.

Typically, codomain of R is subset of reals. A random variable is used kind of like a variable, but it is “implemented” as a function.

Coin example

We flip 3 fair coins. Let C be the random variable that is the number of coins that come up heads. Let M be a random variable that is 1 if all three coins come up heads or all three coins come up tails and 0 otherwise. They are random variables in that they map all possible outcomes to values, integers in this case.

Example: $C(THH) = 2$. $M(THH) = 0$. $C(TTT) = 0$. $M(TTT) = 1$.

C is *counting* the number of heads, M tells us whether or not all the coins *match*.

In terms of sample space

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$C(HHH) = 3 \quad C(THH) = 2$$

$$C(HHT) = 2 \quad C(THT) = 1$$

$$C(HTH) = 2 \quad C(TTH) = 1$$

$$C(HTT) = 1 \quad C(TTT) = 0$$

$$M(HHH) = 1 \quad M(THH) = 0$$

$$M(HHT) = 0 \quad M(THT) = 0$$

$$M(HTH) = 0 \quad M(TTH) = 0$$

$$M(HTT) = 0 \quad M(TTT) = 1$$

Definition

Definition: An *indicator random variable* is a random variable that maps every outcome to either 0 or 1. Indicator random variables are also called *Bernoulli variables*.

Example: The random variable M . It “indicates” whether the three coins match.

Connection between indicator random variables and events. Recall, an event is a subset of the sample space—a set of outcomes. An indicator random variable can be interpreted as a set, since it maps each outcome to whether it is *in* the set (1) or *out* of the set (0).

If E is an event, we can define the corresponding indicator random variable I_E , where $I_E(\omega) = 1$ if $\omega \in E$ and 0 otherwise.

Example: If we take E to be the event where all 3 coins match, $M = I_E$.

Partitioning sample space

An indicator random variable partitions sample space:

$$\underbrace{HHT \ HTH \ HTT \ THH \ THT \ TTH}_{M=0} \quad \underbrace{HHH \ TTT}_{M=1}$$

So does any other random variable:

$$\underbrace{TTT}_{C=0} \quad \underbrace{HTT \ THT \ TTH}_{C=1} \quad \underbrace{HHT \ HTH \ THH}_{C=2} \quad \underbrace{HHH}_{C=3}$$

Statements about random variables

Each block is a subset of the sample space and therefore an event.

The assertion that $C = 2$ defines an event: $\{THH, HTH, HHT\}$.

$$\Pr[C = 2] = 3/8.$$

$$\Pr[M = 1] = 1/4.$$

Statements about random variables can also be viewed as events.

$$\Pr[C \leq 1] = 1/2.$$

$$\Pr[M \cdot C \text{ is odd}] = 1/8.$$

This last statement is a funny way of saying “all heads”. Why?

Concept

The *expected value* (often *expectation*) of a random variable is its mean or probability weighted average.

Example: Define a random variable R to be the alphabetic position of the first letter of the outcome of a coin flip, $R(H) = 8$, $R(T) = 20$. The expected value of R is 14. It is $1/2 \times 8 + 1/2 \times 20$.

We write $\mathbb{E}[R] = 14$. (Book uses “Ex”, but I can’t pretend that’s ever used.)

Suppose we select a student uniformly at random from the class, and let R be the student’s homework 2 score. Then, $\mathbb{E}[R]$ is just the class average. The expected value is a useful thing to know.

Definition

Definition: If R is a random variable defined on a sample space $\mathcal{S}$, then the expectation of R is

$$\mathbb{E}[R] ::= \sum_{\omega \in \mathcal{S}} R(\omega) \Pr[\omega].$$

Example: $\mathbb{E}[C] = \frac{0+1+1+1+2+2+2+3}{8} = 3/2.$

Example: $\mathbb{E}[M] = \frac{1+0+0+0+0+0+0+1}{8} = 1/4.$

Exercise for the reader: If E is an event, $\Pr[E] = \mathbb{E}[I_E].$

Fair die

Let R be the random variable corresponding to a fair die. Here, the outcomes are numbers, so we'll just define $R(\omega) = \omega$.

$$\mathbb{E}[R] = \frac{1+2+3+4+5+6}{6} = 7/2 \text{ or } 3.5.$$

Does that mean we expect the die to come up 3.5? No, it will *never* come up 3.5. Maybe “expected value” was a bad choice of name.

In general, if R is a random variable with a uniform distribution over $[1, n]$, $\mathbb{E}[R] = \sum_{i=1}^n i \cdot \frac{1}{n} = \frac{1}{n} \sum_{i=1}^n i = \frac{1}{n} n(n+1)/2 = (n+1)/2$.

Nonuniform distribution

A cup of coffee has around 95 mg of caffeine. Tea: 26 mg. Hot chocolate: 0 mg.

60% of orders at the CS22 Cafe are for coffee. 25% for tea. 15% for hot chocolate.

What's the expected caffeine content in an order?

$.6 \cdot 95 + .25 \cdot 26 + .15 \cdot 0 = 63.5$ mg: a *weighted* average.

One and die

Let R again be the random variable corresponding to a fair die.

$$1 + \mathbb{E}[R] = 1 + 3.5 = 4.5.$$

$$\mathbb{E}[1 + R] = \frac{2+3+4+5+6+7}{6} = \frac{27}{6} = \frac{9}{2} \text{ or } 4.5.$$

Sometimes the expectation of a function matches the function of the expectation.

One over die

Let R again be the random variable corresponding to a fair die.

$$\frac{1}{\mathbb{E}[R]} = \frac{1}{3.5} = \frac{2}{7} \text{ (.29 ish).}$$

$$\mathbb{E}\left[\frac{1}{R}\right] = \frac{1+1/2+1/3+1/4+1/5+1/6}{6} = \frac{49}{120} \text{ (.41 ish).}$$

Sometimes the expectation of a function matches the function of the expectation.
Sometimes not.